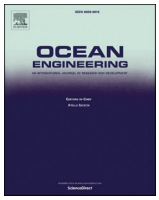
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Research paper



Interaction between ice cover and floating platform simulated by dilated-polyhedron-based DEM

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ABSTRACT

This study presents a dilated-polyhedron-based discrete element method (DPDEM) aimed at investigating the interaction between the self-propelled ice-resistant platform (IRSPP) and drifting sea ice. The IRSPP is treated as a rigid body anchored by a mooring system, providing six degrees of freedom (6-DOF) in translations and rotations. The simulation of the ice-IRSPP interaction utilizes the DPDEM, incorporating contact forces and bond-failure criteria to account for sea ice collisions and fractures, respectively. To address disparities between DEM sea ice and actual sea ice, as well as the random nature of fracture and fragmentation following IRSPP interaction, an analysis of the dimensional independence of DEM calculation elements is conducted. Additionally, the validity of DEM parameters is assessed by comparing the numerical ice resistance of various previous icebreakers with the Lindqvist formula. Then, the time history of structural ice forces, ice heeling moments, and platform motion responses under varying ice floe thicknesses are investigated. Furthermore, the influences of sea ice conditions on ice load and the value of the ice heeling moment for the IRSPP are analyzed.

1. Introduction

The accelerated melting of Arctic sea ice due to climate warming has extended polar research activities, intensifying global interest in polar scientific exploration, resource exploitation, and shipping routes, garnering significant international attention (Ebinger and Zambetakis, 2009; Svistunov et al., 2019). Consequently, the construction of polar drifting research vessels that enable all-terrain access to the Arctic and support year-round scientific research operations becomes of paramount scientific importance. There is a dearth of studies on the ice load experienced by Ice-Resistant Self-Propelled Platforms (IRSPPs) in ice regions. Svistunov (2019) conducted an experimental study in the AARI ice tank to examine the stability of IRSPP under intense ice pressure conditions, providing a schematic diagram illustrating the lateral inclination of the model due to the interaction with sea ice on both sides (Fig. 1).

Furthermore, Maksimova et al. (2021) assessed the ice loads in the mid-body region of the IRSPP and the size and distribution of sea ice along the sides and bottom, utilizing the data derived from the aforementioned model tests (Svistunov et al., 2019). Based on these findings, they established the design of an ice load monitoring system for IRSPPs during ice region drift. Additionally, Zhou et al. (2013) conducted a model experimental study on the icebreaking tanker Uikku to examine the ice-hull interaction under various heading angles of polar ships. They investigated sea ice damage patterns and accumulation phenomena at different sea ice drift angles, observing variations in ice breakup accumulation under different interaction speeds at the same heading angle. Aksnes et al. (2010) conducted an experimental model study on the dynamic response of moored ships subjected to ice loads in level ice. However, their investigation solely focused on the ice-hull interaction in the bow direction, neglecting a comprehensive analysis of the ice-heeling moment resulting from hull-sea ice interaction. While model tests are commonly regarded as the most effective means of examining the interaction between ice and hulls, their results cannot be directly applied in design processes due to potential disparities between the properties of model ice and the target ice (Zhou et al., 2018).

The semi-empirical equation approach and numerical calculation approach both require a range of sea ice parameters as model inputs, including ice thickness, ice concentration, physical properties of the ice (such as bending strength, Poisson's ratio, and elastic modulus), and the ice-hull dynamic friction coefficient (Wang et al., 2021). However, the semi-empirical formula method imposes strict criteria on the choice of operating conditions and hull structure parameter information.

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Additionally, there is currently no empirical formula that meets the prerequisites for calculating bilateral ice force loading on the platform structure caused by drifting ice floes. In conclusion, the numerical calculation method offers significant advantages over field testing, model tests, and semi-empirical formula methods in terms of parameter repeatability, flexibility in defining working conditions, efficiency, and cost-effectiveness. Consequently, the use of numerical methods will facilitate advancements in preliminary research and practical design of polar ships and marine engineering structures. Ni et al. (2020) examined ship resistance in level ice using CFD-DEM (Sun et al. (2023); Zhong et al. (2023); Xue et al. (2024)). Polar ship ice-breaking operations involve intricate uncertainties related to factors such as the ship's attitude, spatial line load distribution, sea ice damage patterns, and the distribution of accumulated broken ice (Su, et al., 2010; Tan et al., 2013; Liu, et al., 2021). A well-constructed numerical calculation model of ice-ship interaction will be a crucial tool for studying ice loads and the anti-ice performance of ships operating in actual ice regions. Currently, various methods such as the Finite Element Method (FEM) (Kärnä, et al., 1999; Liu et al., 2011; Chen et al., 2022), Discrete Element Method (DEM) (Liu, et al., 2018; 2021; DI et al., 2018; Liu et al., 2022), Circumferential Cracking Method (Su, et al., 2010; Tan et al., 2013; Zhou et al., 2018), and Smoothed Particle Hydrodynamics (SPH) method (Chen, et al., 2022) are employed for accurate analysis of the ice-hull interaction process.

Chen et al. (2022) utilized the FEM-SPH adaptive method to construct a numerical model investigating the interaction between an icebreaker and level ice at various heading angles. They assessed the influence of the heading angle on the accumulation of broken ice and ice-breaking forces; however, a comprehensive analysis of the hull's dynamic response was not conducted. Given the nonlinear nature of the ice-breaking process, it is crucial to account for the non-constant ice--breaking process influenced by environmental loads. This involves considering the structural ice loads and their kinematic response within the ship's six-degree-of-freedom (6-DOF) state. Zhou et al. (2018) carried out 6-DOF ice-hull interaction simulations under ice drift reversal conditions. They employed the circumferential cracking method to analyze the structural ice loading and motion response when the ice floe struck the hull at approximately a 180° counterclockwise angle. The study yielded results for the hull's surge, sway, and roll. The DEM has the capacity to simulate the macroscopic-scale fracture and damage of sea ice during ship-ice interaction. Additionally, it can effectively replicate various forms of sea ice, including level ice, ice floes, ice ridges, and hummocked ice (Gong, et al., 2019; DI et al., 2018; Xue et al., 2020; Polojaervi et al., 2012). Quinton (2006) pioneered hull-turning simulations using DECICE discrete element software for model tests of the Terry Fox icebreaker. Subsequently, Di et al. (2018) and Liu et al. (2022) conducted ship-turning calculations of ice-breaking performance in 6-DOF utilizing spherical and polyhedral discrete element methods, respectively. However, few studies have addressed the evaluation of ice resistance while considering bilateral ice pressure effects.

This study aims to address this issue by presenting an enhanced method to simulate the unique scenario of the full-size platform experiencing ice pressure from both sides. Construction is underway for a novel IRSPP designed specifically for polar expeditions. With the aid of an icebreaker, this new research vessel is capable of navigating through Arctic ice floes for a duration of 9 months. The work presented in this study is part of the verification of the ice resistance of this new research vessel.

2. Dilated-polyhedron-based DEM for ice-IRSPP interaction

This study employs the DPDEM code developed by the Polar Ship and Ocean Engineering Group at Dalian University of Technology. Prior to its application, the code underwent rigorous validation through field ice mechanics tests, model tests, and the utilization of semi-empirical formulations, as demonstrated by Liu et al. (2021). To facilitate inter-model contact search, contact force, and hydrodynamic calculations, the dilated polyhedral elements and hull model surfaces are discretized into triangular meshes (Metrikin, 2014). This section introduces a contact model that describes the interaction between level ice and IRSPP, along with a bond-failure criterion, and also provides an explanation of the dynamic calculations for both sea ice and platform structures.

2.1. Dilated-polyhedron-based DEM for sea ice

The dilated polyhedron is generated by merging a smooth spherical spatial geometry point set $\boldsymbol{A}$ with a polyhedral point set $\boldsymbol{B}$ through the principles of Minkowski sum theory. The Minkowski sums of point sets $\boldsymbol{A}$ and $\boldsymbol{B}$ in spatial geometry are defined as (Liu and Ji, 2018):

$$\boldsymbol{A} \oplus \boldsymbol{B} = \{\vec{x} + \vec{y} | \vec{x} \in \boldsymbol{A}, \vec{y} \in \boldsymbol{B}\} \tag{1}$$

where $\vec{x}$ and $\vec{y}$ denote geometric points in the spatial geometry point sets $\boldsymbol{A}$ and $\boldsymbol{B}$, respectively. Here, $\boldsymbol{A}$ and $\boldsymbol{B}$ correspond to the dilated sphere and polyhedron, respectively. The external surface of polyhedron $\boldsymbol{B}$ undergoes a sweeping action by a dilated sphere $\boldsymbol{A}$, resulting in a new dilated polyhedron as the Minkowski sum of dilated sphere $\boldsymbol{A}$ and polyhedron $\boldsymbol{B}$.

The Dilated-Polyhedron-Based Discrete Element Method (DEM), functioning as a block discrete element, offers an enhanced representation of real sea ice scenarios when compared to models based on spherical discrete elements (Liu, et al., 2021). It facilitates a reasonable characterization of sea ice dynamics, encompassing drift transport, overturning, stacking, and sliding. To replicate arbitrary morphological features of natural sea ice, a Voronoi cutting algorithm was employed to parametrically divide a two-dimensional plane into polygons. These polygons were then vertically stretched to yield polyhedral elements, and the Dilated-Polyhedron-Based DEM for sea ice elements was derived based on the Minkowski sum principle (Liu and Ji, 2018), as shown in Fig. 2.

In numerical simulations utilizing the discrete element method, the dynamic interactions among elements involve translational and rotational motions governed by Newton's second law of motion. In this study, the governing equations are applied to the dilated polyhedral

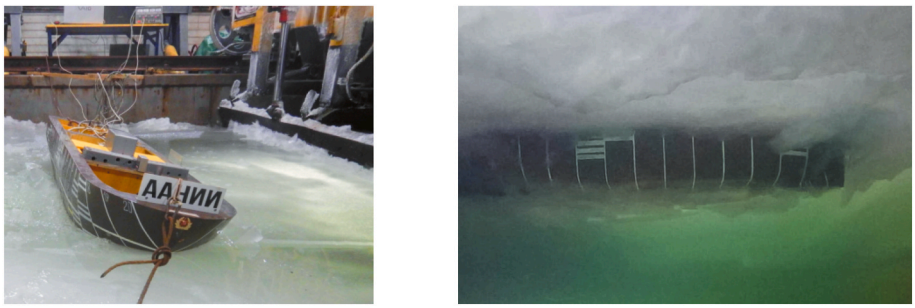


Fig. 1. Fragments from experiments assessing the ice heeling moment (Svistunov, 2019).

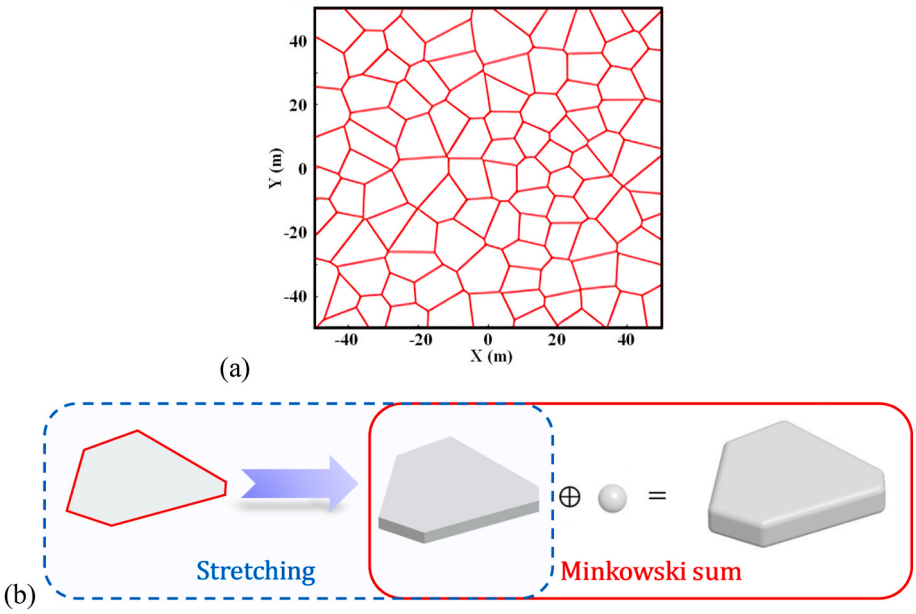


Fig. 2. The generation process of dilated polyhedral sea ice elements: (a) random polygons generated using the Voronoi cutting algorithm; (b) dilated polyhedral sea ice element models created through the Minkowski sum algorithm.

element, expressed as Liu et al. (2022):

$$m_i\frac{d\boldsymbol{v}_i}{dt}=\sum_{j=1}^{N_c}\left(\boldsymbol{F}_{n,ij}+\boldsymbol{F}_{t,ij}\right)+m_i\boldsymbol{g} \tag{2}$$

$$\boldsymbol{I}_i\frac{d\boldsymbol{\omega}_i}{dt}=\sum_{j=1}^{N_c}\left(\boldsymbol{M}_{n,ij}+\boldsymbol{M}_{t,ij}\right) \tag{3}$$

where m represents the mass of a sea ice element. $\boldsymbol{v}$ and $\boldsymbol{\omega}$ denote the translational and rotational velocities, respectively. The subscript i signifies the index of the sea ice element. $\mathbf{g}$ is the acceleration of gravity, and $\boldsymbol{I}$ is the inertia tensor. N_c is the overall number of particles in contact with element i. $\boldsymbol{F}_{n,ij}$ and $\boldsymbol{F}_{t,ij}$ represent the normal and tangential forces between elements i and j, respectively. $\boldsymbol{M}_{n,ij}$ and $\boldsymbol{M}_{t,ij}$ denote the moments of $\boldsymbol{F}_{n,ij}$ and $\boldsymbol{F}_{t,ij}$ acting on element i, respectively.

It is noteworthy that the rotation of the dilated-polyhedron-based DEM for sea ice in the global coordinate system typically induces a change in the inertia tensor. Consequently, each particle possesses a respective local coordinate system, and the quaternion method is employed to determine the transformation relationship between the global and local coordinate systems. The quaternion $\boldsymbol{Q}$ is expressed as $\boldsymbol{Q}=(q_0,q_1,q_2,q_3)^{\mathrm{T}}$, and the corresponding transformation matrix $\boldsymbol{R}$ is expressed as (Kosenko, 1998):

$$\boldsymbol{R}=\begin{bmatrix} q_0^2+q_1^2-q_2^2-q_3^2 & 2(q_1q_2+q_0q_3) & 2(q_1q_3-q_0q_2) \\ 2(q_1q_2-q_0q_3) & q_0^2-q_1^2+q_2^2-q_3^2 & 2(q_2q_3+q_0q_1) \\ 2(q_1q_3+q_0q_2) & 2(q_2q_3-q_0q_1) & q_0^2-q_1^2-q_2^2+q_3^2 \end{bmatrix} \tag{4}$$

The vector $(\boldsymbol{v}^l)$ in the local coordinate system and the vector $(\boldsymbol{v}^g)$ in the global coordinate system conform to the relationship: $\boldsymbol{v}^l=\boldsymbol{R}\bullet\boldsymbol{v}^g$. The moment $(\boldsymbol{M}_{ij}^l)$ acting on element i in the local coordinate system can be expressed as:

$$\boldsymbol{M}_{ij}^l=\boldsymbol{R}\bullet\boldsymbol{M}_{ij}^g \tag{5}$$

where $\boldsymbol{M}_{ij}^g$ represents the moment acting on element i in the global coordinate system. The rotational motion of element i in the local coordinate system can be expressed as:

$$I_{xx,i}^l\frac{\mathrm{d}\omega_{x,i}^l}{\mathrm{d}t}+\left(I_{zz,i}^l-I_{yy,i}^l\right)\omega_{y,i}^l\omega_{z,i}^l=M_{x,ij}^l \tag{6}$$

$$I_{yy,i}^l\frac{\mathrm{d}\omega_{y,i}^l}{\mathrm{d}t}+\left(I_{xx,i}^l-I_{zz,i}^l\right)\omega_{z,i}^l\omega_{x,i}^l=M_{y,ij}^l \tag{7}$$

$$I_{zz,i}^l\frac{\mathrm{d}\omega_{z,i}^l}{\mathrm{d}t}+\left(I_{yy,i}^l-I_{xx,i}^l\right)\omega_{x,i}^l\omega_{y,i}^l=M_{z,ij}^l \tag{8}$$

where the subscripts x, y, and z represent the vector components along the x, y, and z directions. The rotational angular velocity $\boldsymbol{\omega}_i^l$ in the local coordinate system can be determined by solving Eqs. (6)–(8). For a detailed procedure for calculating the quaternion, refer to the work of (Fritzer, 2001).

Two distinct methods are available for contact detection between dilated polyhedrons. One approach involves direct geometric calculation, known as geometry-based contact detection. Dilated polyhedral surfaces encompass spheres, cylinders, and planes, allowing for contact detection between these shapes through geometric computation (Galindo-Torres et al., 2012). This method enhances computational efficiency. An alternative approach involves contact detection based on an approximate envelope function (Liu and Ji, 2020). The approximate envelope function comprises a weighted sum of both polyhedral and spherical functions (Boon et al., 2013).

This study employs contact search algorithms based on approximate envelope functions to compute contact overlap and contact forces between elements of a dilated polyhedron. During element contact interaction, a non-linear contact force model is used to simplify the calculation of inter-element contact forces; in other words, the energy loss due to element collision is also calculated in addition to the elastic recovery force caused by elastic deformation. As a result, the normal contact force $(\boldsymbol{F}_{\mathrm{n}})$ considers both elastic and viscous forces, and the normal and tangential contact forces $(\boldsymbol{F}_{\mathrm{n}})$ and $(\boldsymbol{F}_{\mathrm{t}})$ can be expressed as (Liu and Ji, 2021; Liu et al., 2022):

$$\boldsymbol{F}_{\mathrm{n}}=\boldsymbol{F}_{\mathrm{n}}^{\mathrm{e}}+\boldsymbol{F}_{\mathrm{n}}^{\mathrm{v}}=k_{\mathrm{n}}\delta_{\mathrm{n}}^{\frac{3}{2}}+c_{\mathrm{n}}\sqrt{\delta_{\mathrm{n}}}\dot{\delta}_{\mathrm{n}} \tag{9}$$

$$\boldsymbol{F}_{\mathrm{t}}=\boldsymbol{F}_{\mathrm{t}}^{\mathrm{e}}+\boldsymbol{F}_{\mathrm{t}}^{\mathrm{v}}=\mu\left|\boldsymbol{F}_{\mathrm{n}}^{\mathrm{e}}\right|\left[1-\left(1-\frac{\boldsymbol{\delta}_{\mathrm{t}}}{\boldsymbol{\delta}_{\mathrm{t}}^{\max}}\right)^{3/2}\right]\frac{\boldsymbol{\delta}_{\mathrm{t}}}{\left|\boldsymbol{\delta}_{\mathrm{t}}\right|}+c_{\mathrm{t}}\sqrt{\frac{6\mu m_{\mathrm{AB}}\left|\boldsymbol{F}_{\mathrm{n}}^{\mathrm{e}}\right|\sqrt{1-\boldsymbol{\delta}_{\mathrm{t}}/\boldsymbol{\delta}_{\mathrm{t}}^{\max}}}{\boldsymbol{\delta}_{\mathrm{t}}^{\max}}}\dot{\boldsymbol{\delta}}_{\mathrm{t}} \tag{10}$$

where $\boldsymbol{F}_{\mathrm{n}}^{\mathrm{e}}$ and $\boldsymbol{F}_{\mathrm{n}}^{\mathrm{v}}$ represent the normal elastic force and viscous force, respectively, while $\boldsymbol{F}_{\mathrm{t}}^{\mathrm{e}}$ and $\boldsymbol{F}_{\mathrm{t}}^{\mathrm{v}}$ denote the tangential elastic force and viscous force, respectively. k_{n} is the normal contact stiffness, calculated as $k_{\mathrm{n}}=4/3E^{*}\sqrt{R^{*}}$, where $E^{*}=\frac{E}{2(1-\nu^{2})}$, $R^{*}=\frac{R_i \bullet R_j}{R_i+R_j}$. E is elastic modulus of the sea ice element, ν is the Poisson's ratio, R is the radius of the particle's volume-equivalent sphere. $\boldsymbol{\delta}_{\mathrm{n}}$, $\boldsymbol{\delta}_{\mathrm{t}}$, $\dot{\boldsymbol{\delta}}_{\mathrm{n}}$, and $\dot{\boldsymbol{\delta}}_{\mathrm{t}}$ are the normal and tangential overlap and relative velocities, respectively. c_{n} and c_{t} are the normal and tangential damping coefficients, respectively. μ is the friction coefficient, $\boldsymbol{\delta}_{\mathrm{t}}^{\max}$ is the maximum tangential overlap, calculated as $\boldsymbol{\delta}_{\mathrm{t}}^{\max}=\frac{\mu\boldsymbol{\delta}_{\mathrm{n}}(2-\nu)}{(2-2\nu)}$, and m_{AB} is the equivalent mass, given by $1/m_{\mathrm{AB}}=1/m_{\mathrm{A}}+1/m_{\mathrm{B}}$.

To account for the freezing state of real sea ice under natural conditions, bonding points are established between elements, and adjacent co-planar elements are bonded to each other to characterize the freezing state of sea ice. The bonding force model is shown in Fig. 3. The elastic stresses on the two bonded surfaces are determined based on the elastic matrix σ^{e}. Considering the energy loss in the floating process of sea ice, the normal damping coefficient C_{n} and tangential damping coefficient C_{s} are employed to represent the viscous process. Consequently, the elastic stress σ^{e} between the elements, the viscous stress σ^{v}, and the bond force $\boldsymbol{F}^{\mathrm{b}}$ can be expressed as:

$$\boldsymbol{\sigma}^{e}=\begin{bmatrix}k_n & 0\\ 0 & k_s\end{bmatrix}\boldsymbol{\varepsilon}=\frac{E(1-\nu)}{(1+\nu)(1-2\nu)}\begin{bmatrix}1 & 0\\ 0 & \frac{1-2\nu}{2(1-\nu)}\end{bmatrix}\boldsymbol{\varepsilon} \tag{11}$$

$$\boldsymbol{\sigma}^{v}=\begin{bmatrix}C_n & 0\\ 0 & C_s\end{bmatrix}\dot{\boldsymbol{\varepsilon}}=\beta\begin{bmatrix}k_n & 0\\ 0 & k_s\end{bmatrix}\dot{\boldsymbol{\varepsilon}} \tag{12}$$

$$\boldsymbol{F}^{b}=(\boldsymbol{\sigma}^{e}+\boldsymbol{\sigma}^{v})\bullet\frac{A}{n} \tag{13}$$

where $\boldsymbol{\varepsilon}$ denotes the normal and tangential strains between the bond nodes, given as $\varepsilon=\{\varepsilon_{\mathrm{n}},\varepsilon_{\mathrm{t}}\}^{\mathrm{T}}$; k_{s} is the tangential contact stiffness; β is the constant coefficient, and the damping coefficient is calculated by the constant coefficient β of stiffness; $\dot{\boldsymbol{\varepsilon}}$ represents the strain rate, defined as $\dot{\boldsymbol{\varepsilon}}=\{\dot{\varepsilon}_{\mathrm{n}},\dot{\varepsilon}_{\mathrm{t}}\}^{\mathrm{T}}$; A is the area of the common face; n is the number of bond nodes at the intersection.

In assessing the fracture-breaking dynamics of sea ice subjected to the interaction with structures, a hybrid fracture energy model is employed to ascertain the bonding state of sea ice elements, determining critical strength by concurrently considering normal tensile and shear forces. Following the exceedance of critical stress strength, a softening process in stiffness is incorporated. Subsequently, the fracture energy, bonding state, and critical deformation are computed based on this hybrid fracture energy model. Upon surpassing the critical deformation threshold, the bond connecting sea ice elements fails, leading to the fracture and breakup of the sea ice elements (Liu and Ji, 2021).

For each pair of bond points, the critical elasticity state is determined by a strength determination criterion that factors in both tensile and shear forces, expressed as:

$$\left(\frac{\langle\sigma\rangle}{\bar{\sigma}}\right)^{2}+\left(\frac{\tau}{\bar{\tau}}\right)^{2}\geq 1 \tag{14}$$

where $\bar{\sigma}$ and $\bar{\tau}$ represent the normal and tangential bond strengths, respectively, and $\langle\ \rangle$ denotes the Macaulay bracket ($\langle x\rangle=x$ for $x\geq 0$, $\langle x\rangle=0$ for $x<0$). σ and τ are the normal and shear stress, respectively.

The shear strength is governed by the Mohr-Coulomb law, expressed as:

$$\bar{\tau}=C-\mu_{\mathrm{b}}\sigma \tag{15}$$

where C represents the internal cohesion, and μ_{b} is the coefficient of internal friction, and σ is positive in tension.

As shown in Fig. 4, the inherent connection between equivalent force and deformation is characterized using the bilinear model. The variable describing the degree of material damage is denoted as D, and the stiffness (k') of the material in the damage phase is expressed as:

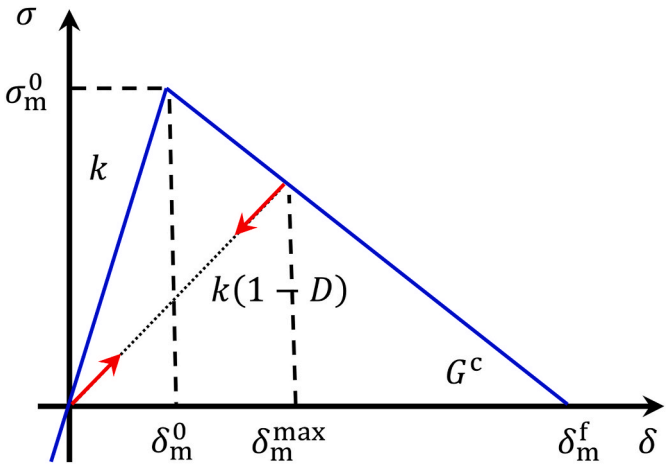


Fig. 4. Bilinear constitutive relationship between equivalent stress and deformation.

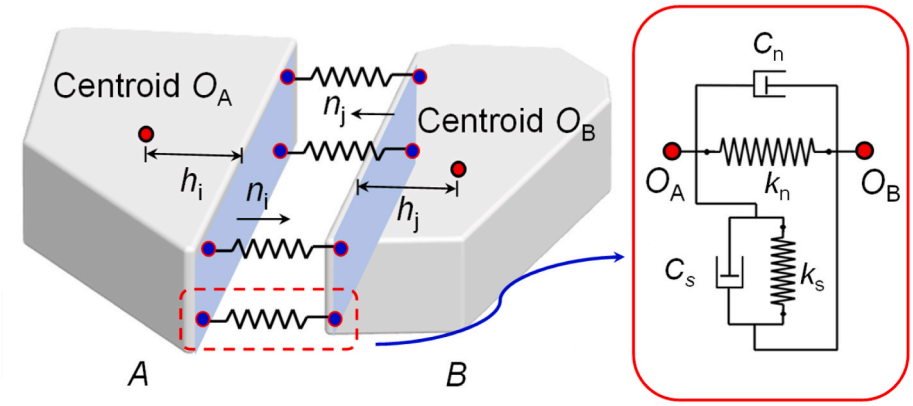


Fig. 3. Bond point settings on bonding surfaces.

$$D = \frac{\delta_{\mathrm{m}}^{\mathrm{f}}\left(\delta_{\mathrm{m}}^{\max} - \delta_{\mathrm{m}}^{0}\right)}{\delta_{\mathrm{m}}^{\max}\left(\delta_{\mathrm{m}}^{\mathrm{f}} - \delta_{\mathrm{m}}^{0}\right)} \tag{16}$$

$$k^{'} = (1 - D)k \tag{17}$$

where $\delta_m^{\max}$ is the maximum deformation at the bonding point during the loading history, and δ_m^{f} is the critical deformation at bond failure. In other words, bond failure occurs when δ_{m} exceeds δ_m^{f}. The calculation of δ_m^{f} is derived from the equation $\delta_{\mathrm{m}}^{\mathrm{f}} = 2G^{\mathrm{c}}/\sigma_{\mathrm{m}}^{0}$, where G^{c} is the critical hybrid fracture energy, the fracture energy is the same as those considered by Paavilainen and Tuhkuri (2012). The Benzeggagh-Kenane model is employed to determine this hybrid fracture energy, and it can be expressed as (Benzeggagh and Kenane, 1996):

$$G^{\mathrm{c}} = G_{\mathrm{I}}^{\mathrm{c}} + \left(G_{\mathrm{II}}^{\mathrm{c}} - G_{\mathrm{I}}^{\mathrm{c}}\right)\left(\frac{G_{\mathrm{II}}}{G_{\mathrm{I}} + G_{\mathrm{II}}}\right)^{\eta} \tag{18}$$

where $G_{\mathrm{I}}^{\mathrm{c}}$ and $G_{\mathrm{II}}^{\mathrm{c}}$ represent the critical fracture energy for tensile and shear-type cracks, respectively. Additionally, η is a constant factor, $\eta = 1.75$ for brittle materials (Camanho et al., 2003; Liu and Ji, 2021). The current state tensile and shear-type fracture energies, G_{I} and G_{II}, can be calculated from the deformation in tension and shear as a ratio of the two, as expressed (Liu and Ji, 2021):

$$\frac{G_{\mathrm{II}}}{G_{\mathrm{I}}} = \frac{\delta_{\mathrm{s}}^{2}}{\langle \delta_{\mathrm{n}} \rangle^{2}} \tag{19}$$

2.2. Numerical calculation of simulation setup

This section introduces a computational model that incorporates the influences of ice forces and hydrodynamics on the hull, aiming to analyze the 6-DOF motion response of the IRSPP during sea-ice compression. Employing this approach, a discrete element simulation is executed to investigate the interaction between the platform model and sea ice. Initially, the hull structure's surface is discretized into several triangular elements, facilitating contact search and force calculation between the numerical sea ice elements and the hull structure (Liu and Ji, 2018). Ice loads acting between the sea ice element and the hull triangular element are computed using a non-linear contact model based on Hertz's theory. The normal force is derived from the normal stiffness and overlap, while the tangential force is incrementally calculated following the Mohr-Coulomb law (Liu and Ji, 2018).

Moreover, the IRSPP exhibits translational movement along the overall coordinate system and rotation within the coordinate system of the ship, responding to external loads to achieve a 6-DOF motion. The motion information is updated using the quaternion method, as specified in Eq. (4). Fig. 5 shows the coordinate system for the motion response of the platform. The motion equation of the platform can be expressed as (Liu and Ji, 2022).:

$$\boldsymbol{F}_i + \boldsymbol{F}_b + \boldsymbol{F}_d + \boldsymbol{F}_p + \boldsymbol{F}_{\mathrm{g}} = m\frac{d\mathbf{u}}{dt} \tag{20}$$

$$\boldsymbol{M}_i + \boldsymbol{M}_b + \boldsymbol{M}_d + \boldsymbol{M}_p = \boldsymbol{I}\frac{d\omega}{dt} \tag{21}$$

where m and $\boldsymbol{F}_{\mathrm{g}}$ represent the mass and gravity of the platform; $\boldsymbol{I}$ is the inertia tensor of the dilated polyhedron. $\mathbf{u}$ and ω represent the increments of displacement and angle in a single time step, respectively. The subscripts i, b, d, and p indicate the ice force, buoyancy force, drag force, and the force of the mooring system or thruster along with its respective moment.

The impacts of seawater buoyancy, current drag forces, and their respective moments on the 6-DOF system of the ship are taken into account through a discretization approach. The buoyancy is computed utilizing Archimedes' principle, expressed as (Liu and Ji, 2018):

$$\boldsymbol{F}_b = -\rho_w V_{sub}\mathbf{g} \tag{22}$$

where ρ_{w} represents the density of seawater; $\mathbf{g}$ denotes the acceleration of gravity; V_{sub} is the total volume of the hull submerged in water.

Given that the hull surface is composed of triangular elements, an equivalent tetrahedron is used to calculate the submerged volume when the hull is immersed in water. The center of gravity of the element is denoted as O_{p}. If face ABC is entirely submerged, a tetrahedron $O_{\mathrm{p}}ABC$ is formed, as shown in Fig. 6. The tetrahedral center of mass ($\boldsymbol{x}_i$) and the submerged volume (V_i) can be determined. If face ABC intersects the water line, the intersection vertices E and F are calculated, and the corner points on the water surface with E and F are replaced. Then, the center of mass and volume of the tetrahedron formed by the submerged point and E and F are calculated. In this scenario, if two vertices cross the waterline, the tetrahedron $O_{\mathrm{p}}AEF$ is obtained, as shown in Fig. 7(a). If a vertex crosses the waterline, then the face $ABFE$ will be divided into two triangles (ABF and AEF), forming the tetrahedra O_pABF and O_pAEF, as shown in Fig. 7(b).

Considering a single submerged tetrahedron O_pABC as an example, its element center of mass, $\boldsymbol{x}_i$, and submerged volume, V_i, can be expressed as (Metrikin, 2014):

$$\boldsymbol{x}_i = \frac{1}{4}\left(\boldsymbol{x}_{O_p} + \boldsymbol{x}_A + \boldsymbol{x}_B + \boldsymbol{x}_C\right) \tag{23}$$

$$V_i = \frac{1}{6}h \bullet \overrightarrow{AB} \bullet \overrightarrow{BC} \tag{24}$$

where h is the perpendicular distance from point O_{p} to face ABC. It is important to note that the right-hand spiral rule should be satisfied, ensuring that the normal is directed outward. The total submerged

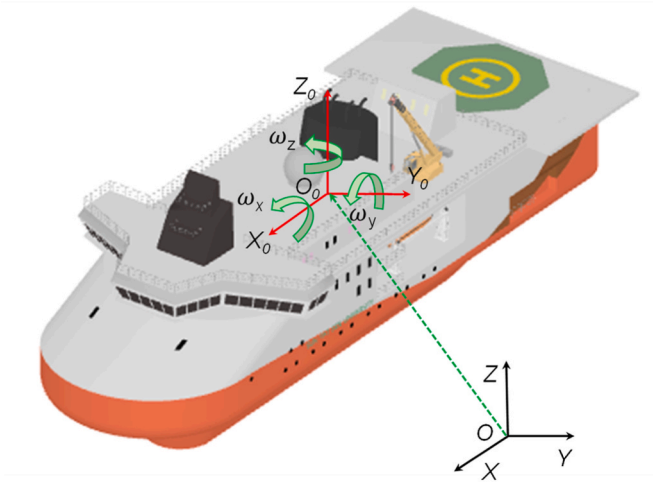


Fig. 5. The navigational coordinate system of IRSPP.

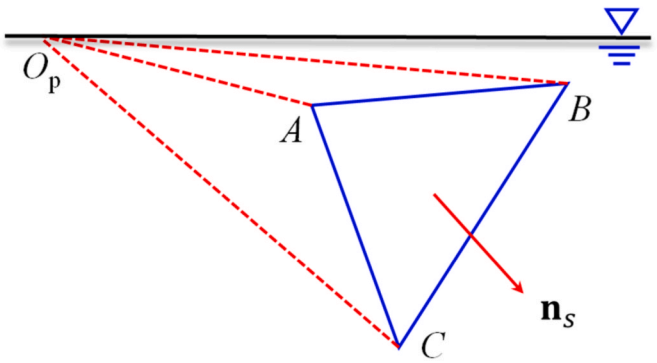
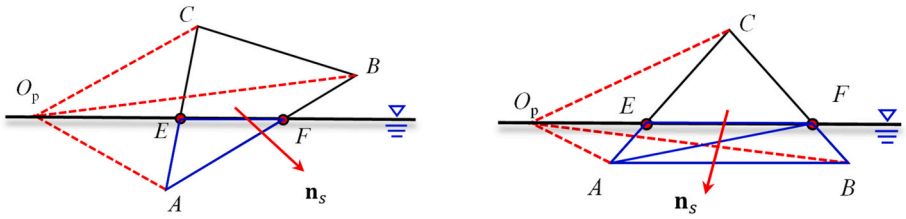


Fig. 6. Surface ABC fully submerged in water.



(a) Two vertices across the waterline (b) One vertex across the waterline

Fig. 7. Surface ABC intersecting the waterline.

volume of the floating structure (V_{sub}) and the center of buoyancy coordinate of the floating structure ($\boldsymbol{x}$) can then be expressed as:

$$V_{\text{sub}} = \sum V_i \tag{25}$$

$$\boldsymbol{x} = \frac{\sum \boldsymbol{x}_i V_i}{V_{\text{sub}}} \tag{26}$$

Then, the buoyancy force $\boldsymbol{F}_{\text{b}}$ and buoyancy moment $\boldsymbol{M}_{\text{b}}$ can be calculated. The buoyancy moment is a result of the imbalance between the center of buoyancy and the center of gravity, potentially leading to the overturning of the floating body structure (Liu and Ji, 2018). The buoyant moment generated by the buoyancy force on the center of mass of a ship can be expressed as:

$$\boldsymbol{M}_{\text{b}} = (\boldsymbol{x} - \boldsymbol{x}_{\text{struct}}) \times \boldsymbol{F}_{\text{b}} \tag{27}$$

where $\boldsymbol{x}_{\text{struct}}$ is the center of gravity of the floating body structure in the global coordinate system, and $\boldsymbol{x} - \boldsymbol{x}_{\text{struct}}$ is the distance from the center of gravity to the center of buoyancy.

The buoyant and drag forces on the hull from the water are calculated due to the interaction with currents, while the interactive force on the water from the element is not considered. Both hull drag forces and moments can be calculated by summing the contributions from individual triangular elements (Liu and Ji, 2018; Sun and Shen, 2012). This can be expressed as:

$$\boldsymbol{F}_d = -0.5 C_d^F \rho_w \sum_{i=1}^{N} A_{sub}^i (\boldsymbol{v}_s - \boldsymbol{v}_w) |\boldsymbol{v}_s - \boldsymbol{v}_w| \tag{28}$$

$$\boldsymbol{M}_d = -0.5 C_d^M \rho_w \sum_{i=1}^{N} A_{sub}^i r_{eff}^3 \boldsymbol{\omega} |\boldsymbol{\omega}| + \boldsymbol{r}_d^i \times \boldsymbol{F}_d^i \tag{29}$$

where N is the number of divided triangular surfaces; C_d^F is the drag force coefficient of the triangular element on the ship surface; ρ_w is the density of seawater; A_{sub}^i is the wet surface area; $\boldsymbol{v}_s$ is the velocity of motion at the center of the triangular surface; $\boldsymbol{v}_w$ is the flow velocity; C_d^M is the drag moment coefficient of the triangular element; r_{eff} is the effective radius of the submerged hull; $\boldsymbol{r}_d^i$ is the distance vector from the center of gravity to the center of each triangular element.

3. Verification of dilated-polyhedron-based DEM

Due to the confidentiality of the IRSPP data, the ice resistance simulation of the R/V *Xuelong* for comparison with the Lindqvist formula is used to validate the accuracy of the numerical model presented in this study. Subsequently, simulation calculations under the ice region operating conditions of the polar drifting station were conducted. The Lindqvist formulas referenced in Liu et al. (2021) were employed for this analysis.

3.1. Simulation condition

Full-scale models of the R/V *Xuelong* and IRSPP were employed in the DEM simulations, where the vessel mesh was partitioned into triangular elements. This division facilitated the calculation of the inertia tensor of the hull, assessment of the fluid forces acting on it, and detection of the state of ice-vessel contact, as shown in Fig. 8. The key parameters of the ice cover utilized in the DEM simulation are detailed in Table 1 (Liu and Ji, 2021), and the structural dimensions of R/V *Xuelong* and IRSPP are provided in Table 2.

To streamline the numerical simulation process, the displacement constraints were imposed on the sea ice model boundary. It ensures either fixed sea ice or a predefined constant drift velocity to simulate ship-ice interaction. The vessel heading angle (θ) was adjusted to account for relative motion. Here, θ represents the angle between the X-axis and the X_0-axis, as shown in Fig. 9. Among them, Fig. 9(a) shows the working arrangement of the interaction between the ship and unilateral sea ice. The simulation setup of the interaction between the ship and bilateral sea ice will be detailed in the following section 4.1, as shown in Fig. 9 (b).

3.2. Validation of ice load numerical calculation results

The computational element size plays a critical role in influencing simulation outcomes, especially in dilated-polyhedron-based DEM calculations (Huang et al., 2014). This section delves into the previous investigation on element size convergence (Liu and Ji, 2021). Specifically, simulations for ice resistance of the icebreaker are conducted with an element size of 3 m in working conditions. Further details on the test setup can be found in Ref. (Liu and Ji, 2021).

The discrete element method was employed to calculate the time history of ice resistance for the R/V *Xuelong* with a bow angle of 0°, an ice thickness of 0.5 m, and a velocity of 5 m/s, as shown in Fig. 10 (a). In the ice-breaking process of the R/V *Xuelong*, the time of interaction closely correlates with the time history of ice resistance. The region marked as I in Fig. 10(a) indicates that the hull model is not fully embedded in the ice within the initial 40 s and gradually increases with the interaction time. Consequently, this phase is excluded from the ice resistance characteristics analysis. Beyond 40 s of ice-ship interaction time, as denoted by the region II, the ship hull is fully embedded in the ice, and the ice resistance time history stabilizes, leading to ice force fluctuations under a random sea ice break-up mechanism.

Building on this foundation, an analysis of ice resistance characteristics at various ice thicknesses was conducted. This analysis primarily encompasses peak values, means, and peak mean values calculations. However, extracting the peak value of each load individually is deemed inappropriate. Therefore, the method proposed by Qu et al. (2006) for extracting the peak ice force, shown in Fig. 10 (b), was adopted in this study.

The DEM simulation results, presented in Fig. 11 alongside the Lindqvist results, reveal that the Lindqvist outcomes fall between the peak and peak mean values of the DEM-simulated ice resistance. The Lindqvist results exhibit the smallest mean values and demonstrate a non-linear correlation between the peak value and ice thickness. Both the DEM simulation results and the Lindqvist results exhibit an increase in growing ice thickness and display a commendable level of agreement.

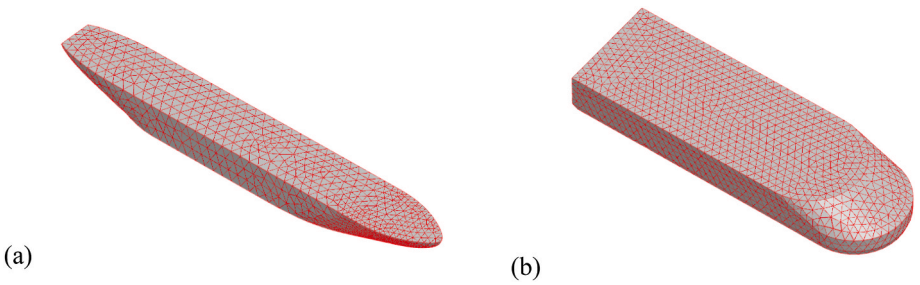


Fig. 8. A computational model of the hull: (a) R/V *Xuelong*; (b) IRSPP.

Table 1
Major computational parameters of ice cover in the DEM simulation.

Definitions	Symbols	Values
Size of level ice	$l \times w$	800 m × 120 m
Seawater density	ρ_w	1035 kg/m^3
Sea ice density	ρ_i	920 kg/m^3
Ice-ice friction coefficient	μ_{ii}	0.1
Ice-hull friction coefficient	μ_{ih}	0.1
Elastic modulus of sea ice	E	2.4 GPa
Normal strength	$\overline{\sigma}$	0.6 MPa
Cohesion strength	C	1.8 MPa
Poisson's ratio of sea ice	ν	0.3
Gravity acceleration	g	9.8 m/s^2
Coefficient of drag force	C_d	0.4
Mode I fracture energy	G_I^c	12 J/m^2
Mode II fracture energy	G_{II}^c	12 J/m^2

Table 2
Major parameters of ship structures in the DEM simulation.

Definitions	Symbols	R/V *Xuelong*	IRSPP
Ship length	L	147.2 m	85 m
Ship breadth	B	22.6 m	22.5 m
Ship draft	T_d	9.0 m	8.5 m
Ship displacement	Δ	21000 t	10000 t
Stem angle	φ	24 °	–
Waterline entrance angle	α	20 °	–

This verifies the validity of the current DEM simulation parameters for accurately calculating ice resistance.

3.3. Convergence analysis of the DEM for ice-IRSPP interactions

In this section, our focus is on simulating interactions between ice floes and a platform with a 90° heading angle. The emphasis is on conducting a convergence analysis of the discrete element sizes for sea ice. Larger element sizes enhance computational efficiency but also influence the accuracy of the platform structure's ice load analysis.

Fig. 12 shows the evolution of drifting sea ice after interacting with the IRSPP structure. The conditions include an ice thickness of 0.5 m, sea ice drift velocity of 0.15 m/s, element size of 2 m, and additional simulation parameters listed in Table 1. We also divided the IRSPP into three areas for analysis and discussion. It can be observed from Fig. 12 (a) that the sea ice approached and contacted one side of the IRSPP, and the speed of numerical sea ice generation changed, but no fracture and fragmentation occurred for the time being. When the simulation reaches 160s, the phenomenon of sea ice fragmentation and accumulation can be found, showing a trend of more accumulation in the mid-body and the accumulation gradually decreases to both ends (Fig. 12(b)). It can be observed from Fig. 12(c) that the accumulated ice begins to displace from the bottom of the mid-body. At this time, the mid-body, bow and stern simultaneously clear the sea ice and form a large ice "cushion " at the bottom of the platform. Finally, the amount of sea ice accumulation reaches its peak, and the process of dynamic equilibrium between accumulation and removal is realized (Fig. 12(d)). This accumulation changes the IRSPP's motions in roll (heeling angle) and draft, thus affecting the stability of the IRSPP. The scenario of sea ice fragmentation and accumulation is consistent with the simulation results of ice fragmentation under the interaction between ship 90° heading angle and level ice carried out by Chen et al. (2022).

Fig. 13 shows the ice resistance dynamics during the interaction of the IRSPP with level ice. These variations exhibit similar characteristics to the ice force results observed in the bow-breaking operation shown in Fig. 10. Both scenarios can be segmented into two phases: an initial rise in ice force followed by a period of stability. It's worth noting that, during the rising phase I the first peak in ice force is succeeded by a steady increase. This pattern contrasts with the consistently upward trend in ice force observed in the bow-breaking operation. The deviation arises because a smaller stem angle can generate a more substantial downward force, demonstrating superior ice-breaking performance that aids in bending and breaking sea ice. However, this results in a smaller contact area, hindering sea ice buildup. As the ship hull embeds deeper into the sea ice, the ice load gradually increases. In contrast, when the mid-body region of the IRSPP engages with drifting sea ice, the interaction area is extensive. This wider structure relative to the sea ice is less conducive to enhanced ice-breaking capacity, necessitating a higher ice-breaking load compared to an icebreaker with a bow structure.

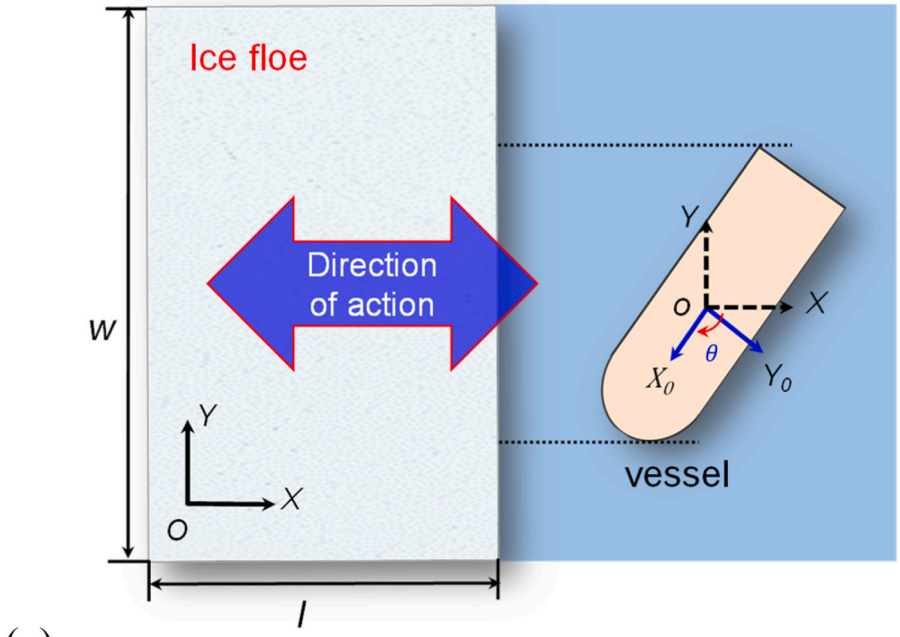
In a similar manner, effective ice loads within the stable region of the IRSPP were isolated for the convergence analysis of element size. As shown in Fig. 14, the ice resistance characteristics at various element sizes, ranging from 1.5 m to 8 m, are presented. This includes the mean values of the raw data (F_m) and mean peak values (F_{mp}). Notably, Fm is consistently smaller than F_{mp}, with F_m/F_{mp} ranging from 60% to 90%, averaging at 83%. The ice resistance exhibits a decreasing trend, reaching a stable range as the element size decreases. Stability in ice resistance is achieved when the element size is less than 3.0 m. Subsequently, for computational efficiency, the element size is set to 2.5 m in the subsequent simulations.

4. Ice load characteristics under platform drift operations

Simulations of the IRSPP, subjected to intense ice pressure on both sides, were conducted utilizing the proposed DPDEM. The parameters related to ice thickness were analyzed to discern the characteristics of ice loads and the dynamic response of the platform structure.

4.1. Simulation setup

In the presence of natural forces, the polar drifting sea ice exerts significant pressure around the platform structure. To replicate this scenario, the simulation setup is shown in Fig. 9 (b). The IRSPP model is situated between two ice floes. The thicker sea ice on the right-hand side is under a fixed constraint, whereas the thinner ice floe on the left is subject to a displacement constraint, allowing it to drift at a constant velocity in a specific direction. Consequently, the ice-breaking event is anticipated to occur from the side of the thinner ice floe, while the



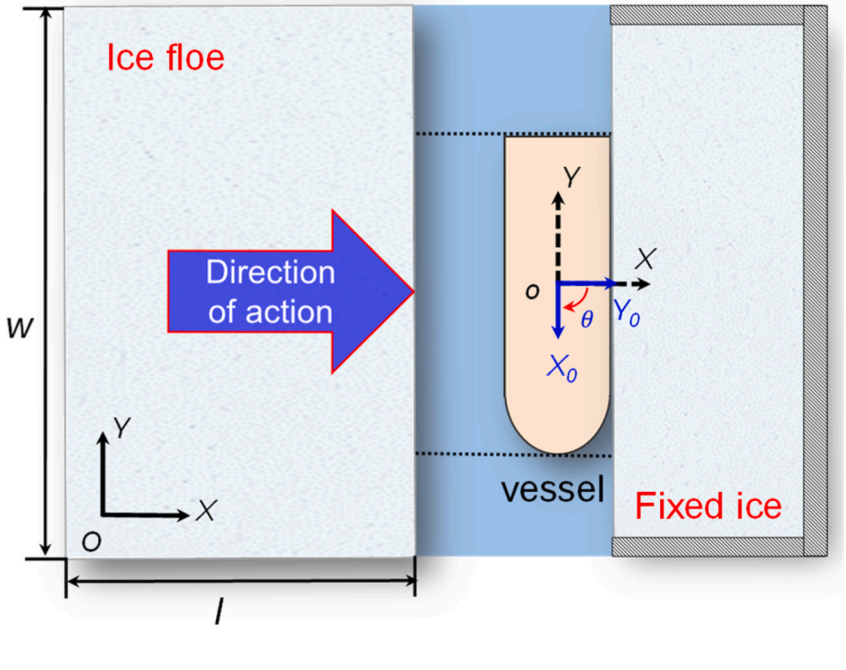


Fig. 9. Configuration of ship operating conditions in the ice region.

platform model remains stationary. The specified conditions involve thin sea ice on the left side, the floating ice is drifting along the *X*-axis, an ice velocity (v_i) of 0.15 m/s, and a range of ice thicknesses (h_i) from 0.5 m to 1.3 m. DEM simulations were carried out to investigate the ice – loading characteristics of the platform structure, taking into account the variations in ice thickness.

Fig. 15 (a) shows the simulation scenario in which the platform is under the action of bilateral sea ice driven by the left ice floe, where the ice floe parameters are set as $h_i = 0.5$ m and $v_i = 0.25$ m/s. Fig. 15 (b) offers a perspective view of ice accumulation in the mid-body region. It is evident that the thinner sea ice slips and fractures along the hull's side upon interaction with the platform, undergoing significant bending. The broken ice initially slides along the side of the thin ice interaction, rises to accumulate, descends to the platform's bottom, forming an ice "cushion," and eventually reaches the side of the thick ice interaction.

Conditions in the stern and bow regions are similar to those in the mid-body. However, the accumulation of broken ice is relatively less pronounced. Influenced by the disturbance from broken sea ice, the IRSPP undergoes a 6-DOF motion response involving heave and roll. This motion persists until the accumulation and discharge of broken ice

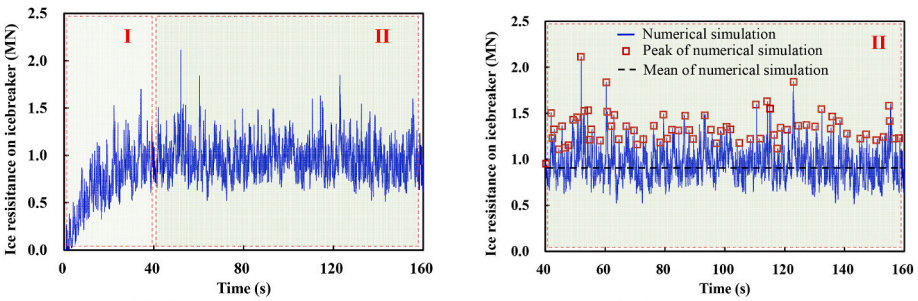


Fig. 10. Time history of ice resistance on the R/V *Xuelong*.

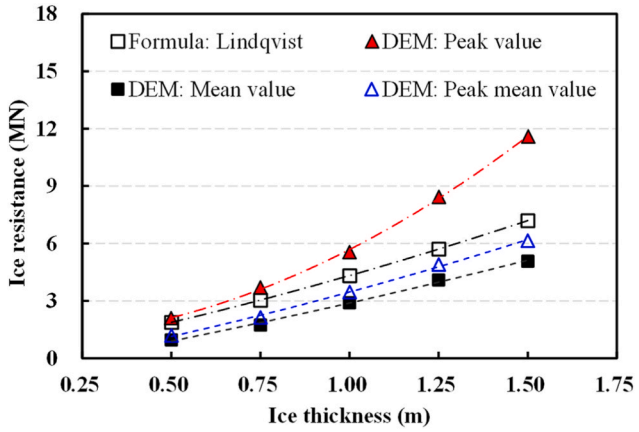


Fig. 11. Relationship between the characteristics of the ice resistance and Lindqvist.

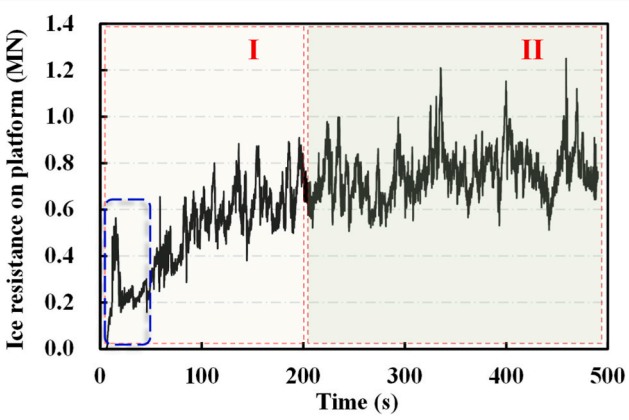


Fig. 13. Time history of ice resistance during the interaction of the IRSPP.

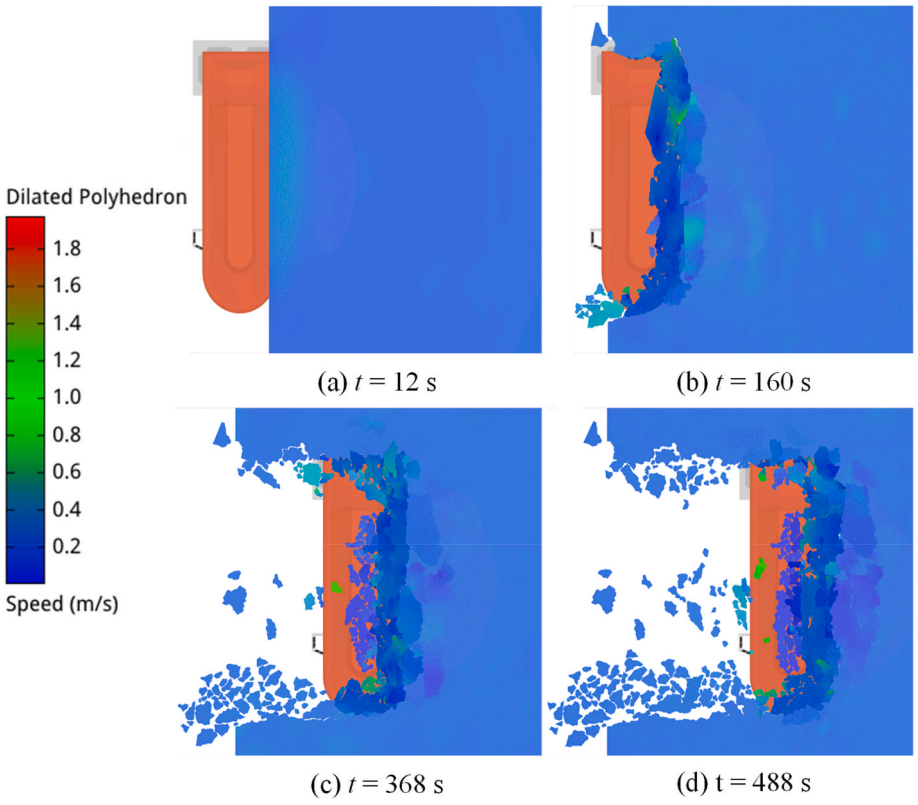


Fig. 12. The process of ice "cushion" formation under the IRSPP.

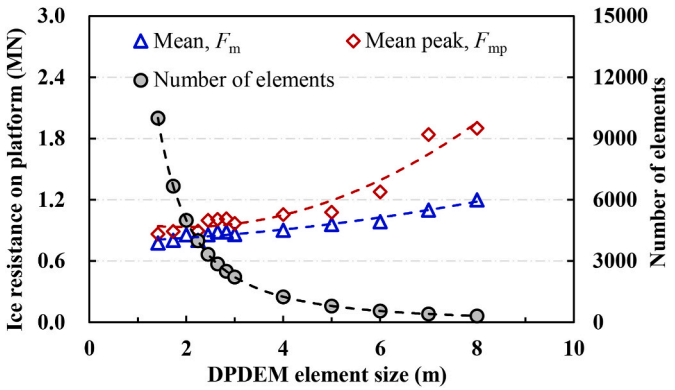


Fig. 14. Relationship between DPDEM element size, IRSPP ice resistance, and the number of elements.

reach a dynamic equilibrium, leading to the ship attaining a stable state. The element size of the ice floe is set at 2.5 m, and the fixed ice thickness is 2.0 m. The velocity of the ice floes is 0.15 m/s. Additional details about the remaining parameters in the DEM simulations are provided in Table 3.

4.2. Simulation results

As an illustration, the simulation Case No. 101 in Table 3, Section 4.1. Fig. 16 presents information on the ice load of the platform and motion response, including the time history of the vertical ice load, heave, ice heeling moment, angular velocity in roll and roll motion under the combined interaction of the ice floe and fixed ice. Fig. 16 (a) reveals that the initial peak in both load and motion response occurs within the first 100 s of the interaction time history, representing the ultimate load generated when the vertical load of the platform exceeds the load-carrying capacity of the sea ice, subsequent platform vertical oscillations and vertical ice force changes tend to flatten out, and the moment of peak occurrence is consistent with the period of action. Driven by floating ice, the platform drifts into contact with thicker fixed sea ice for the first time, and the peak of ice inclination moment is greater than zero in the first 50s (Fig. 16 (b)). Moreover, the absence of significant peaks in the time history of the ice heeling moment during the steady phase suggests that the platform's ice heeling moment is primarily induced by buoyancy and frictional forces from the ice "cushion", and a peak load occurs in the case of the non-simultaneous break-up of the thinner ice floe edge, followed by a decline.

At the same time, the IRSPP shows a rolling motion response under the ice force on both sides, and its angular velocity in roll is shown in Fig. 16 (c). The angular velocity in roll oscillates continuously around 0 rad/s, and the response is not significant, which is due to the unilateral thick ice fixed constraint and unbroken, which makes the platform effectively constrained by the motion. Integration of the angular velocity in roll time course yields the roll shown in Fig. 16 (d), where it can be observed that the platform is generally tilted towards the side of the ice floe, with a peak roll of 0.1° in the stable region (500 s–600 s), which is in agreement with the modelled kinematic response law obtained from the ice flume model tests of Svistunov et al. (2019), as shown in Fig. 17 (a). It should be noted that the platform model we developed is different from the Russian model test ship type, and the numerical simulation is a full-scale model. To this end, we only compare the roll degree variation rule of the model test and the numerical results qualitatively, and observe the consistency of the numerical simulation and the model test results. Fig. 17 (b) provides a comparison between the numerical simulation results of the first 350 s of IRSPP motions in roll and the results of the model test. The roll greater than 0° indicates that the platform is tilting toward the fixed ice side, and conversely, the platform is inclined transversely toward the floating ice side. The overall roll amplitude is low, determined by the thickness of the thinner ice floes.

4.3. Discussions

The DPDEM was employed to simulate the ice load and motion response of the platform under various ice floe thicknesses. The structural stability of the platform was evaluated based on both load information and motion response.

4.3.1. Analysis of vertical ice load and heave

Fig. 18 (a) shows the vertical load on the platform under the influence of ice pressure from both sides, encompassing peak values, peak mean values, mean values in the steady phase, and the first peak in the rising phase. The vertical load of the platform structure exhibits an upward trend with increasing ice thickness, mirroring the behavior observed in the bow-breaking mode of the R/V *Xuelong*. For an ice thickness of 1.0 m, the vertical load is lower than that for a thinner ice thickness of 0.9 m. This discrepancy is attributed to the random distribution of the ice "cushion" at the bottom of the platform but doesn't alter

Table 3
Various numerical simulation cases and computational time.

Simulation No.	Ice floe thickness (m)	Simulation time (s)
101	0.5	700
102	0.6	600
103	0.7	600
104	0.8	600
105	0.9	700
106	1.0	500
107	1.1	600
108	1.3	600

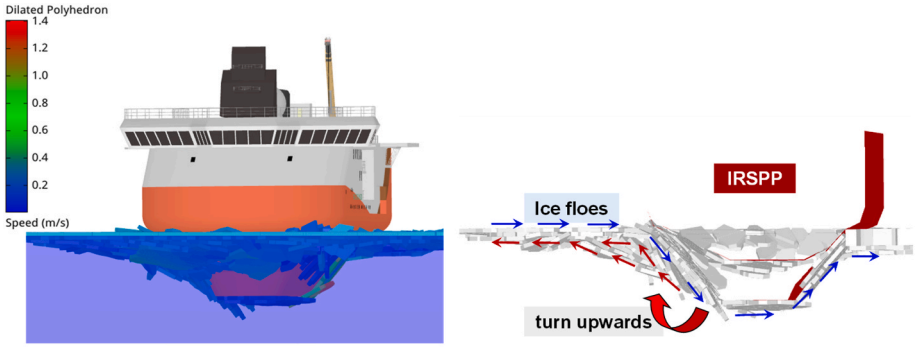
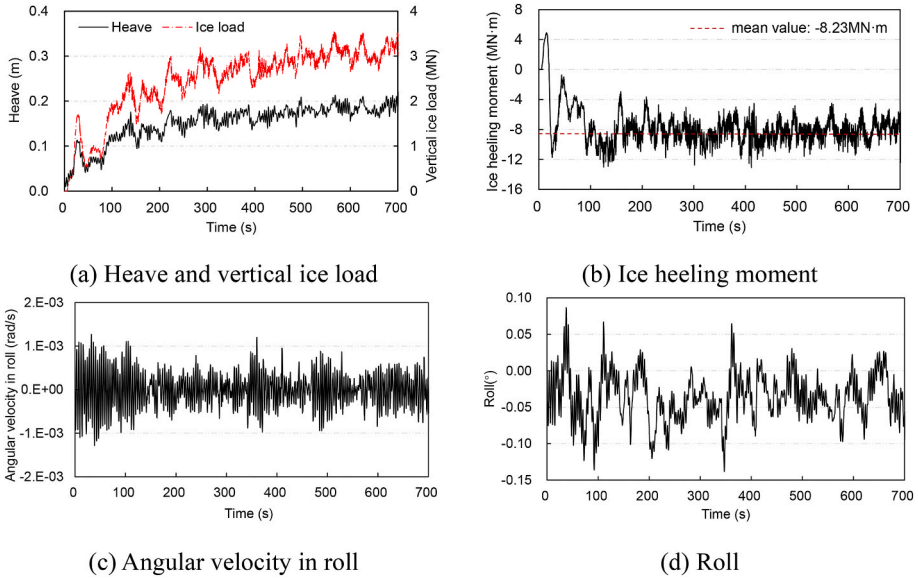


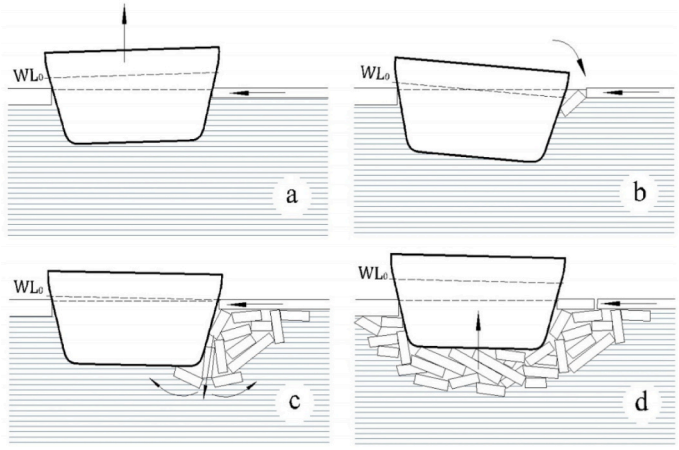
Fig. 15. A schematic of ice rubble and platform interaction: (a) the schematic of the ice-platform interaction; (b) the forming process of an ice "cushion".



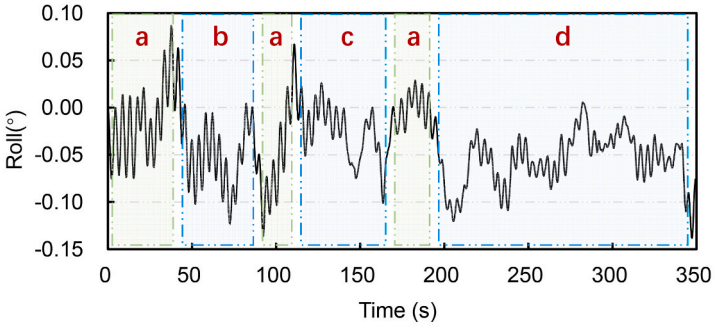
(a) Heave and vertical ice load (b) Ice heeling moment

(c) Angular velocity in roll (d) Roll

Fig. 16. Ice loads and motion response of the IRSPP in Case No. 101.



(a) Main stages of interaction of the platform model with an ice floe at pressures. (Svistunov, et al., 2019).



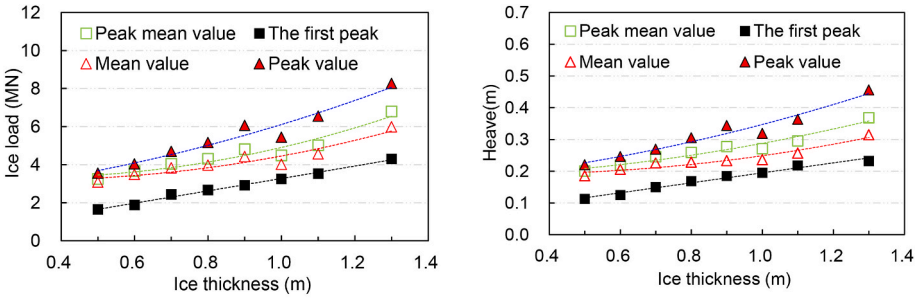
(b) Calculation of IRSPP motions in roll with the DPDEM.

Fig. 17. Time history of the roll motion corresponding to the model tests.

the overall trend. Notably, the pattern of the first peak during the rising phase of the ice force exhibits a strictly positive correlation with ice thickness. Correspondingly, the IRSPP's motions in heave is influenced by the bottom ice "cushion" and the vertical load, aligning with the trend observed in the vertical load, as shown in Fig. 18 (b).

4.3.2. Analysis of ice heeling moment

Fig. 19 shows the ice heeling moment of the platform under varying ice thicknesses, presenting the mean value, peak moment, and peak mean value. For clarity in understanding the load response pattern, all statistics are provided as absolute values, and the interaction direction



(a) Vertical ice loads (b) Heave motion

Fig. 18. Effects of ice thickness on ice load and heave.

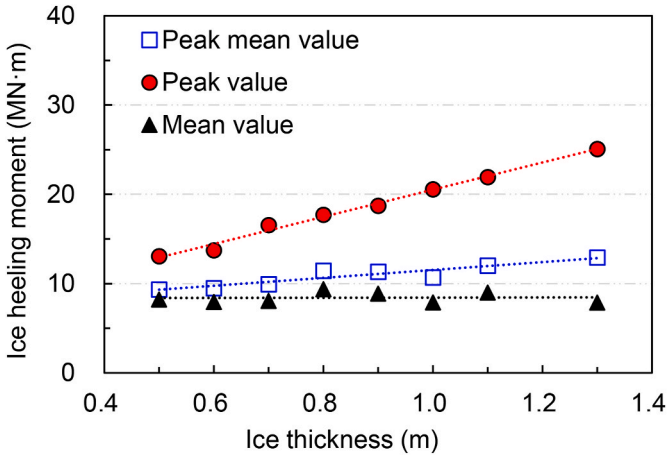


Fig. 19. Influence of ice thickness on ice heeling moment.

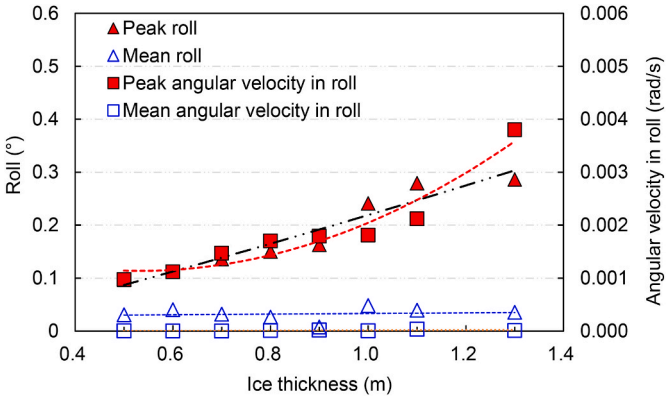


Fig. 20. Influence of ice thickness on roll and angular velocity in roll.

can be inferred from Fig. 17. With an increase in ice floe thickness, more ice force is applied to the structure along the broken side of the platform. Except for the mean value of the ice heeling moment, both the peak and peak mean values increase with the ice thickness.

Upon scrutinizing the time history of the ice heeling moment (Fig. 17 (b)) alongside the discrete element simulation results (Fig. 16), it becomes apparent that thicker sea ice induces a larger vertical component of the load. As the vertical load approaches the load-carrying limit of the ice floe, the ice floe side undergoes a sudden breakup, resulting in a sharp drop in the moment. The minimum ice heeling moment value decreases with increasing ice thickness. Notably, the impact of the random, discontinuous breakup of the ice floes is negligible, as the ice thickness does not influence the mean value of the ice heeling moment, which consistently oscillates around 8000 (kN·m).

4.3.3. *Analysis of roll and angular velocity in roll*

Fig. 20 presents the statistical outcomes of IRSPP motions in roll and angular velocity in roll influenced by ice floe thickness. Nonetheless, the mean roll motion is not sensitive to ice thickness, but is greater than 0°. This is because the sea ice on the side of the ice floe acting on the platform breaks and fails continuously, and the platform tilts to the initial equilibrium position under the action of the restore moment, resulting in the platform tilting to the ice breaking side. Under the joint influence of the sea ice on both sides, the angular velocity in roll and roll motion always fluctuate around 0°. The above periodic response results in a relatively stable mean roll.

5. Conclusions

In this study, a 6-DOF kinematic response and load analysis simulation solver for floating structures are developed. The solver is based on the DPDEM code from the Polar Ship and Marine Engineering Group at Dalian University of Technology, enabling the analysis of ice-IRSPP interactions. Several conclusions can be drawn from the simulation and analysis of various ice-platform interaction scenarios:

During the interaction between ice floes and the platform, the platform's ice force time course exhibits an initial peak during the rising process. This behavior contrasts with the smooth rising trend of ice force in the bow-breaking operation of an icebreaker. The large area of interaction with the ice floes on the platform, characterized by a wider structure compared to sea ice, results in lower ice-breaking capability, necessitating a higher ice-breaking load for the first breakup. The variation pattern of the first peak in the ice force rise phase is strictly positively correlated with ice thickness.

Under the combined action of sea ice on both sides of the platform, the platform's motion response initially tilts toward the fixed ice side. Subsequently, as the floating ice breaks up, the platform tilts toward the floating ice side. This motion is influenced by the vertical component of the ice load and the "cushion" of ice accumulated on the bottom of the platform. The platform eventually enters into a periodic motion response, and the numerical simulation results are in agreement with model test outcomes.

The time profiles of platform motion in heave and roll, vertical ice load, ice heeling moment, and angle velocity in roll were analyzed. Notably, ice thickness is the primary factor influencing vertical ice load, heave, peak ice heeling moment, peak roll, as well as peak angle velocity in roll. All these parameters increase with ice thickness. However, the mean ice tumbling moment and mean transverse rocking angle are not sensitive to ice thickness.

This method provides insights into the dynamics of the platform's initial heave over time due to the effect of the ice "cushion" blocking the bottom and the final value of this change. It's important to note that this study solely focuses on the load and motion response of ice interacting with the platform and does not consider fluid-solid coupling. The flow

field resulting from the platform's bottom motion significantly affects the motion of the ice "cushion" and the ice loads. The next step involves investigating ice-platform interaction using a coupled CFD-DEM numerical simulation method based on computational fluid dynamics methods.

CRediT authorship contribution statement

Xiang Wang: Writing – original draft, Methodology, Investigation. **Lu Liu:** Supervision, Investigation. **Siqiang Wang:** Supervision, Resources. **Shunying Ji:** Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper

Data availability

Data will be made available on request.

Acknowledgments

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